AI-Based Student Performance Prediction System :

➢ Problem Statement:

Educational institutions often struggle to detect students who may underperform in the upcoming exam.

Without early identification ,teachers cannot give extra help on time. Current methods are reactive rather than proactive. This project aims to develop a predictive model that uses historical and behavioral data to forecast student outcomes early in the semester

➢ Why this problem matters?

Students academic performance is the key indicator of educational success and institutional effectiveness. However , many educational institutions face challenges in identifying student who may struggle academically before their result decline significantly. Traditional monitoring methods often detect the problems only after the poor performance already occurred.

An AI- based student performance prediction system can address this challenge by analyzing historical academic data and identifying the patterns that influence future performance.

➢ Domain Selection :

- Education Technology (EdTech)

➢ Target Variables:

- The expected outcome is the Student's expected GPA ,grade, or percentage in the next semester

Problem scoping steps

1. Understand the business problem

-Initial statement :

"The university has observed variation in the academic performance"

- Need to uncover the actual problem :

The AI team will ask themselves multiple questions:-

To academic department-

-What is the current performance rate ?

- Since when has it increased ?

- which departments are effected?

To student welfare-

-What interventions currently exists?

-How much time is available before interventions?

To management-

- why you think the problem id important?
- what impact does failure have socially and financially ?

**Suppose the university reverts the following response-

- current performance rate = 50%
- Desired performance rate = 90-100%
- Most Academic variations occurs during the 2nd year
- counselors can support around 150 students each semester.
- early identification can improve retention.

2. Define the problem precisely

- **Poor problem statement:**

“ predict whether the student will fail ”

-Better approach:

“ Identify second year student at high risk of underperforming in the next semester at least one semester before failing , so that counselors or teachers can intervene ”

- **Who ?** second year students
- **What?** Predict the filing risk
- **When?** One semester before
- **Why?** So that counselors can intervene
- **Outcome** Improve the Academic performance

3. Is AI really needed ?

- **can the problem we solve without AI?**
- **AI should only be used when :**

-patterns are complex

-Multiple factors interact

-Manual rules are insufficient

- **In our case:**

-Academic performance depends on many factors.

-Attendance , study hours, family conditions, health condition,

Student engagement, previous semester trends…

4. Identifying stake holders

Student – receive timely support

Counselors- to identify student needing help

University management – reduce failures

Parents – student success

Faculty – Improve student outcome

Department – integrate and maintain

University – their reputations

Society – successful youth

5. Define the AI task

-Machine learning formulation:

- **classification of problem**:
  -output = fail-> YES | fail->no
  (binary classification)
- **Risk scoring** :
  - how likely is failure = 0-100%
- **ranking problem** :
  - which 150 students should counselors contact?

6. Defining the success metrics

- **Accuracy**
- **Precision**
- **Recall**
- **ROC-AUC**

7. Defining the constraints

- types of constrains

- **Operational constraints :**
  - counselors can only support 150 student/ semester
- **Data constraints :**

-Certain information may not exist

-mental health indicators

-family circumstances

❖ **Ethical constraints :**

-data privacy and Security

-psychological impact and stigmatization

-Algorithmic bias and fairness

8. Assumptions and risk

-Assumption:

-historical data reflects the future trend.

-Attendance records are accurate.

- counselors act on the recommendations

-students are giving their honest review

-Risk :

- Missing data.
- Biased historical decision.
- unexpected events that effect the failure .
- changes in the academic policies.

Methodology -

